POLARIS

Spring Research Project: Mentor Rubric

Mentor Name(s): __ [redacted] _____

Mentee Name(s): __ [redacted] _____

Project Title: _____Subhalo Orbits in Cold and Self-Interacting Dark Matter Halos_____

In a short paragraph, explain the project and its learning objectives:

Dark matter halos form the backbone of cosmic structure, and the orbital motion of subhalos inside larger host halos plays a major role in galaxy evolution. In Cold Dark Matter (CDM), host halos follow well-understood "NFW" profiles, while Self-Interacting Dark Matter (SIDM) can significantly modify halo density profiles over time, producing cores and later gravothermal collapse. In this project, the mentee will learn the physics of CDM and SIDM, develop intuition for how gravitational potentials shape orbits, and implement orbit integration in Python using both analytic NFW potentials and realistic host halo profiles extracted from SIDM simulations. They will learn to translate physical models into code, work with astrophysics Python packages such as *galpy*, and create clear, visually appealing scientific plots that compare the behavior of orbits under different physical assumptions.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

- A working Python script or notebook that integrates subhalo orbits in:
  - an analytic NFW potential
  - a realistic host halo potential constructed from simulation-derived M(<r)
- 3D orbit visualizations for both potentials
- Plots of orbital radius vs. time, velocity vs. time, and comparisons between models
- (optional) Energy vs. time and angular momentum vs. time
- (optional) Radial phase-space plot (r, v_r)
- (optional) Apocenter/pericenter and orbital period comparisons

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Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/16) Intro to project	Reading: introduction to dark matter, large-scale structure, CDM, and SIDM motivation. Coding: install Python, Jupyter, numpy, matplotlib, galpy; run a simple Python script to confirm environment works; learn about how to generate data and save in txt files.
Week 2 (2/23) In class work day	Reading: • NFW halo profiles (basic formula and interpretation) • SIDM small-scale challenges (cores vs cusps, too-big-to-fail) Coding: learn about plotting (2D+3D); make a plot for predicted NFW density profile.
Week 3 (3/2) In class work day	Reading: learn about how cosmological N-body simulations work and how SIDM is implemented Coding: learn to code for a SIDM host halo data table to use for N-body simulations
Week 4 (3/9) Outside of class work	Coding: make 3D orbit plots predicted by NFW
Spring Break	None
Week 5 (3/23) In class work day	Reading: learn about characteristics for halo structure (density profile, velocity dispersion profile, enclosed mass profile, Jean equation, etc) (Mentor will prepare N-body simulation data using the table mentee generated) Coding: make actual 3D orbit from the N-body data

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Week 6 (3/30) Outside of class work	Coding: make 3D orbit plot predicted from realistic host halo data;
Week 7 (4/6) In class work day	Coding: make comparison plots
Week 8 (4/13) Poster work day	Work on poster
Week 9 (4/20) Poster presentation!	None

Reading List: (Make sure material is level appropriate.)

https://arxiv.org/abs/2310.09910
https://www.nature.com/articles/s42254-019-0127-2
Dark matter chapters from Cosmology textbooks

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

Visual Studio Code + Python

Other Notes:

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